Project Proposal: AI-Generated UGC Human Realism Enhancement System

1. Problem Statement

AI-generated User-Generated Content (UGC) is increasingly used for marketing, social media, and advertising. However, even with detailed and carefully engineered prompts, generated videos often have an artificial appearance. Common issues include unnatural facial expressions, fixed eye gaze, robotic body movements, unrealistic blinking, repetitive gestures, poor lip synchronization, and inconsistent motion between frames. These imperfections reduce viewer trust and make the content visibly AI-generated.

The proposed project aims to develop a **Machine Learning-based Human Realism Enhancement System** that automatically evaluates AI-generated UGC videos, identifies the characteristics that make them look artificial, and recommends improvements to increase their human-like appearance.

2. Proposed Solution

The system will act as an intelligent quality-control and optimization layer around an existing AI video-generation system.

Input: User prompt + AI-generated UGC video

Output: Human-realism score + detailed defect analysis + improved prompt/recommendations.

The system will analyze the generated video at both **spatial and temporal levels**. Spatial analysis will examine individual frames, while temporal analysis will study how the person moves across consecutive frames.

3. Implementation Methodology

Step 1 – Video Preprocessing

The generated video will be divided into frames using OpenCV/FFmpeg. Important frames and temporal sequences will be selected for analysis.

Step 2 – Feature Extraction

Computer Vision models will extract features related to:

- Facial landmarks and expressions
- Eye movement and blinking
- Head pose and gaze
- Body posture and gestures

- Walking and arm movement
- Lip movement and synchronization
- Motion consistency between frames
- Hair and clothing movement
- Optical-flow-based motion patterns

Tools such as **OpenCV, MediaPipe and pose/face estimation models** can be used.

Step 3 – Human Realism Model

A labelled dataset containing real human UGC and AI-generated UGC will be created using a structured annotation pipeline. Data will be collected from diverse sources (social media clips, synthetic AI videos, controlled recordings) and standardized in terms of resolution, frame rate, and duration. Human evaluators will assign realism scores using a rubric-based interface (Likert scale + attribute tagging), ensuring consistency across annotators. Inter-annotator agreement (Cohen's Kappa) will be used to validate label reliability.

For feature extraction and modeling, a **multi-stream deep learning architecture** will be used:

- **Visual spatial features:** Extracted using pretrained CNN backbones such as **ResNet-50 / EfficientNet / Vision Transformers (ViT)** to capture facial structure, texture realism, and frame-level artifacts.
- **Temporal dynamics:** Modeled using **LSTM/GRU networks or Temporal Transformers (TimeSformer / Video Swin Transformer)** to capture motion continuity, gesture flow, and temporal consistency across frames.
- **Facial landmark & pose features:** Extracted using tools like **MediaPipe / OpenPose / FaceMesh**, then encoded into structured motion vectors.
- **Audio-visual synchronization (if audio exists):** Modeled using **cross-modal transformers or SyncNet-style architectures** to evaluate lip-sync and speech alignment.

These modalities will be fused using a **late-fusion or attention-based fusion layer**, allowing the model to learn weighted importance across facial, motion, and temporal cues.

The final model will be trained as a **multi-task regression system**, producing an overall **Human Authenticity Score** along with sub-scores for each realism dimension.

Example output:

Overall Realism: 68/100

Component-level predictions:

- Facial realism: 85%
- Eye/gaze realism: 52%
- Body-motion realism: 61%
- Gesture realism: 48%
- Lip synchronization: 90%

- Temporal consistency: 57%

Training will use a combination of **MSE loss (for score regression)** and **ranking loss (to preserve relative realism ordering between samples)**.

Step 4 – Defect Identification

Once the model produces sub-scores, a **diagnostic reasoning layer** will be applied to convert numerical weaknesses into human-interpretable explanations. This will be implemented using a combination of:

- **Feature attribution methods** such as SHAP or Integrated Gradients to identify which visual/temporal features contributed most to low scores
- **Rule-based mapping layer** that links low sub-scores to predefined defect categories
- Optionally, an **LLM-based explanation module** to generate natural language diagnostics from structured outputs

The system will identify key realism breakdowns such as:

> Fixed eye gaze, insufficient natural body movement, repetitive gestures, low micro-expression variance, or temporal jitter between frames.

This step ensures the system is **explainable AI (XAI)-compliant**, making it suitable for iterative content improvement rather than acting as a black-box scorer.

Step 5 – Prompt Optimization

The detected weaknesses will be passed into a **prompt optimization engine powered by a Large Language Model (LLM)**. This module will use structured defect inputs and apply **prompt rewriting strategies such as constraint injection, style augmentation, and behavioral conditioning**.

Techniques used include:

- **Instruction tuning / prompt rewriting with LLMs (GPT-style models)**
- **Reinforcement learning from feedback (RLHF-style loop for prompt quality improvement)**
- **Template-based controlled generation with dynamic slot filling**
- Optional: **embedding similarity search to retrieve high-performing prompt patterns**

The LLM will translate technical defects into natural language generation constraints that directly influence video synthesis models.

For example:

Original Prompt:
“Woman talking about a skincare product.”

Improved Prompt:
“A woman naturally shifts her weight while speaking, uses subtle asymmetric hand gestures, blinks naturally, occasionally shifts her gaze, shows micro facial expressions, and maintains realistic conversational body movement with smooth temporal transitions.”

This ensures that detected realism gaps are explicitly encoded into the next generation cycle.

Step 6 – Iterative Improvement

The system operates as a **closed-loop reinforcement-style optimization pipeline**, where each iteration improves output realism based on feedback signals.

Pipeline flow:

Prompt → Video Generation Model (e.g., diffusion-based video model or GAN-based generator) → Multi-modal ML Analysis → Realism Scoring → Defect Attribution → LLM-based Prompt Refinement → Regeneration

To ensure convergence, the system can include:

- **Stopping criteria based on threshold realism score (e.g., ≥ 85/100)**
- **Early stopping if improvements plateau across iterations**
- **Reward function tracking improvement delta per iteration**

Over multiple cycles, the system effectively behaves like a **self-improving generative feedback loop**, where the generator and evaluator co-evolve, progressively reducing artifacts and increasing human-likeness in UGC-style video content.

4. Expected Outcome

The final system will provide an automated platform capable of:

1. Evaluating how human-like an AI-generated UGC video appears.
2. Identifying specific visual, facial, speech and motion-related defects.
3. Providing a quantitative Human Authenticity Score.
4. Automatically generating targeted prompt improvements.
5. Comparing multiple generated videos and selecting the most realistic version.
6. Reducing manual trial-and-error in prompt engineering.

5. Proposed Technology Stack

Python, OpenCV, MediaPipe, PyTorch/TensorFlow, XGBoost/Random Forest, LSTM/GRU/Transformers, Sentence Transformers/LLMs, FFmpeg, FastAPI and a web-based dashboard.

6. Project Novelty

Instead of attempting to build another large-scale video-generation model, this project focuses on developing an **ML-based intelligence layer that evaluates, explains and improves AI-generated UGC**. The system can potentially work with different video-generation models and provide measurable feedback to the content-generation pipeline.

Final Goal: Develop a practical ML system that helps AI-generated UGC move from *“technically generated”* to *“convincingly human-like.”*